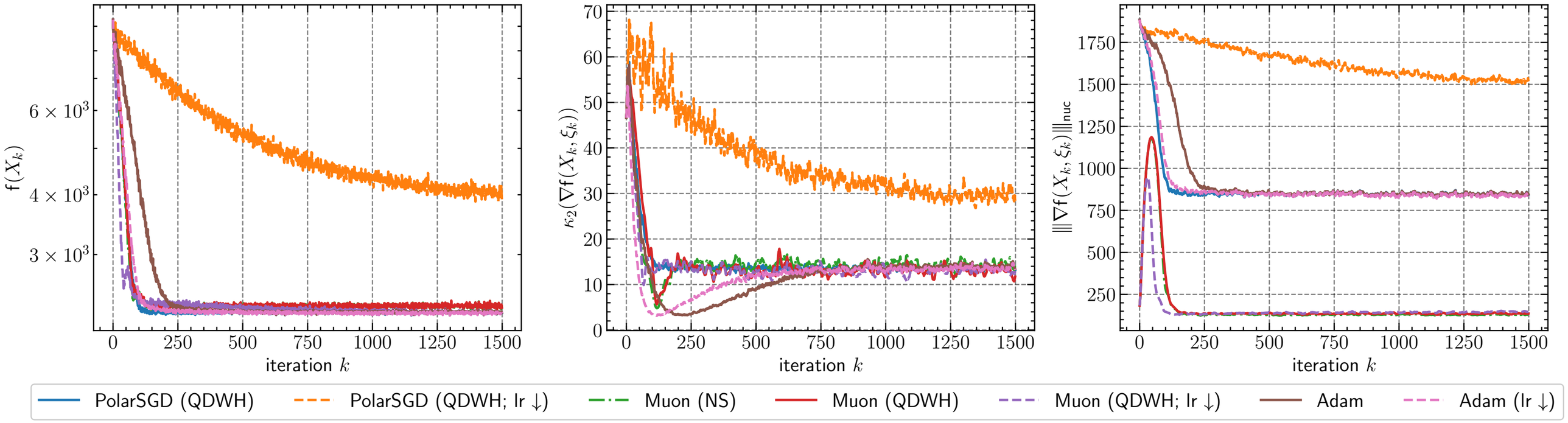
$\mathsf{f}(X_k)$
6×10^3
4×10^3
3×10^3
iteration k
$\kappa_2(\nabla \mathsf{f}(X_k, \xi_k))$
iteration k
$|||\nabla \mathsf{f}(X_k, \xi_k)|||_{\mathsf{nuc}}$
iteration k
PolarSGD (QDWH)
PolarSGD (QDWH; lr ↓)
Muon (NS)
Muon (QDWH)
Muon (QDWH; lr ↓)
Adam
Adam (lr ↓)